BCSE306
Artificial Intelligence

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Objectives

- To impart artificial intelligence principles, techniques and its history.
- To assess the applicability, strengths, and weaknesses of the basic knowledge representation, problem solving, and learning methods in solving engineering problems
- To develop intelligent systems by assembling solutions to concrete computational problems

Course Outcome

1. Evaluate Artificial Intelligence (AI) methods and describe their foundations.
2. Apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation and learning.
3. Demonstrate knowledge of reasoning and knowledge representation for solving real world problems.
4. Analyze and illustrate how search algorithms and planning play vital role in problem solving.

Module:1	**Introduction**	**6 hours**
Introduction- Evolution of AI, State of Art -Different Types of Artificial Intelligence-Applications of AI-Subfields of AI-Intelligent Agents- Structure of Intelligent Agents-Environments		
Module:2	**Problem Solving based on Searching**	**6 hours**
Introduction to Problem Solving by searching Methods-State Space search, Uninformed Search Methods – Uniform Cost Search, Breadth First Search- Depth First Search-Depth-limited search, Iterative deepening depth-first, Informed Search Methods- Best First Search, A* Search		
Module 3	**Local Search and Adversarial Search**	**5 hours**
Local Search algorithms – Hill-climbing search, Simulated annealing, Genetic Algorithm, Adversarial Search: Game Trees and Minimax Evaluation, Elementary two-players games: tic-tac-toe, Minimax with Alpha-Beta Pruning.		
Module:4	**Logic and Reasoning**	**8 hours**
Introduction to Logic and Reasoning -Propositional Logic-First Order Logic-Inference in First Order Logic- Unification, Forward Chaining, Backward Chaining, Resolution.		
Module:5	**Uncertain Knowledge and Reasoning**	**5 hours**
Quantifying Uncertainty- Bayes Rule -Bayesian Belief Network- Approximate Inference in Bayesian networks		
Module:6	**Planning**	**7 hours**
Classical planning, Planning as State-space search, Forward search, backward search, Planning graphs, Hierarchical Planning, Planning and acting in Nondeterministic domains – Sensor-less Planning, Multiagent planning		
Module:7	**Communicating, Perceiving and Acting**	**6 hours**
Communication-Fundamentals of Language -Probabilistic Language Processing -Information Retrieval- Information Extraction-Perception-Image Formation- Object Recognition.		

Text Book	
1.	Russell, S. and Norvig, P. 2015. Artificial Intelligence - A Modern Approach, 3^{rd} Edition, Prentice Hall.

Reference Books	
1.	K. R. Chowdhary, Fundamentals of Artificial Intelligence, Springer, 2020.
2	Alpaydin, E. 2010. Introduction to Machine Learning. 2^{nd} Edition, MIT Press.

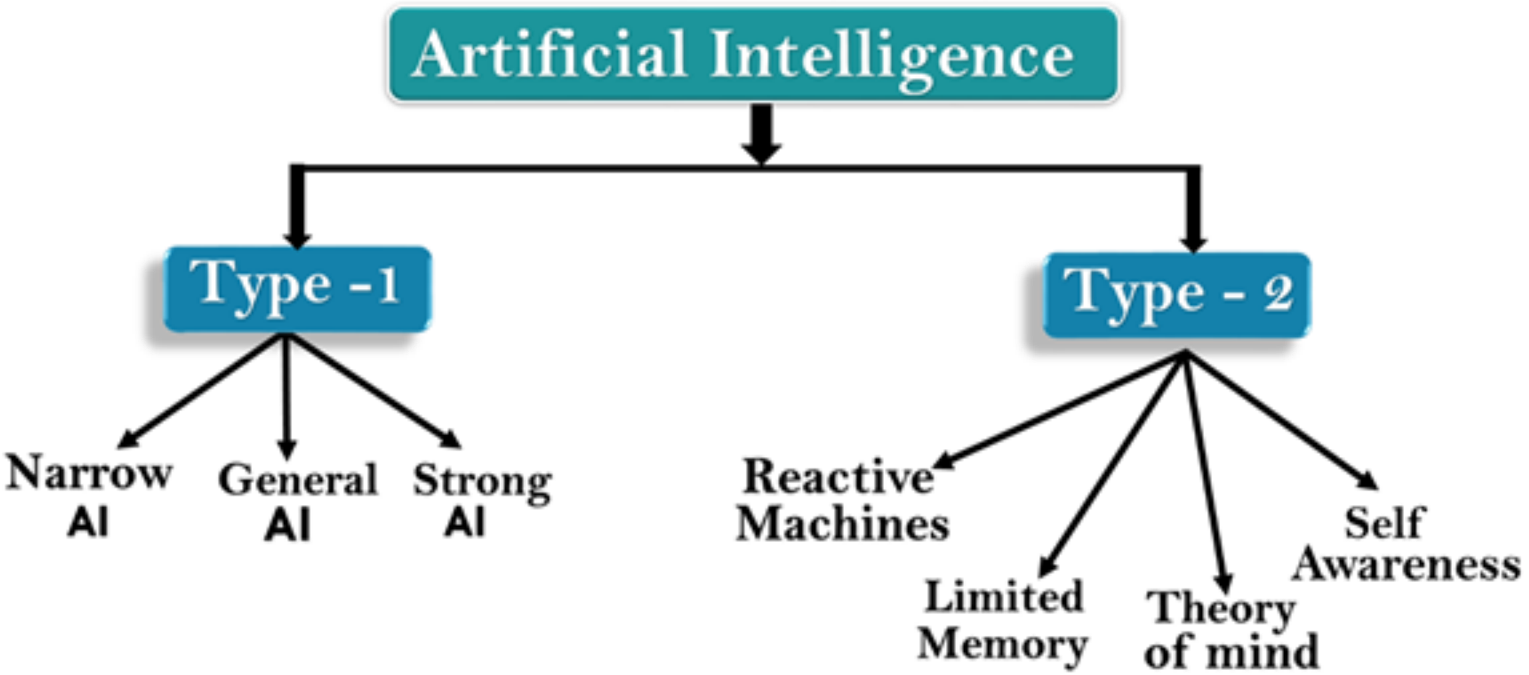
ONLINE COURSE

- https://nptel.ac.in/courses/106106226
- https://onlinecourses.nptel.ac.in/noc23_cs09/preview

Types of Artificial Intelligence

Types of Artificial Intelligence

- Artificial Intelligence can be divided into various types, there are mainly two types of categorization which are based on the capabilities and based on functionality of AI.
- Following is a flow diagram that explains the types of AI.



AI type-1: Based on Capabilities

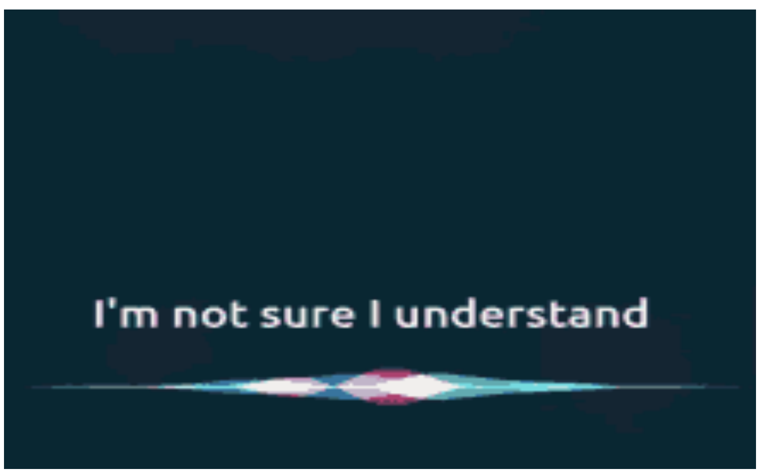
1. Weak AI or Narrow AI (1/3)

- Narrow AI is a type of AI that can perform a dedicated task with intelligence.
- Narrow AI cannot perform beyond its field or limitations, as it is only trained for one specific task. Hence it is also termed as weak AI.
- Narrow AI can fail in unpredictable ways if it goes beyond its limits.

AI type-1: Based on Capabilities

1. Weak AI or Narrow AI (2/3)

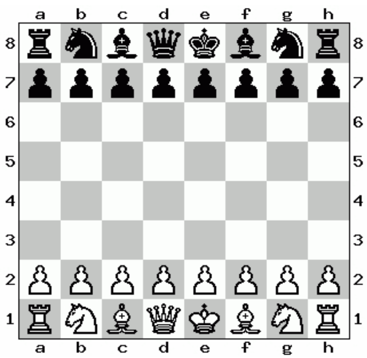
- Apple Siri is a good example of Narrow AI, but it operates with a limited pre-defined range of functions.



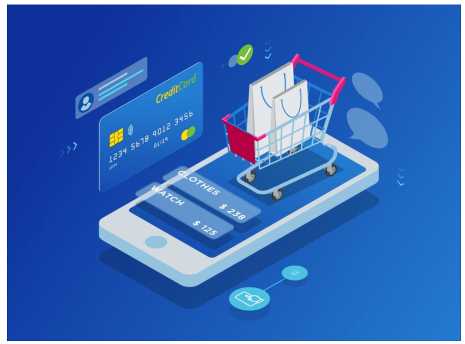
- IBM's Watson supercomputer also comes under Narrow AI, as it uses an Expert system approach combined with Machine learning and natural language processing.

AI type-1: Based on Capabilities

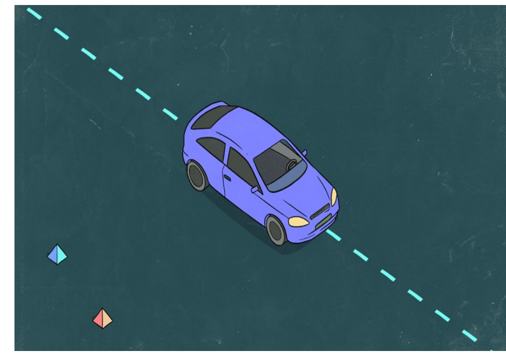
1. Weak AI or Narrow AI (3/3)



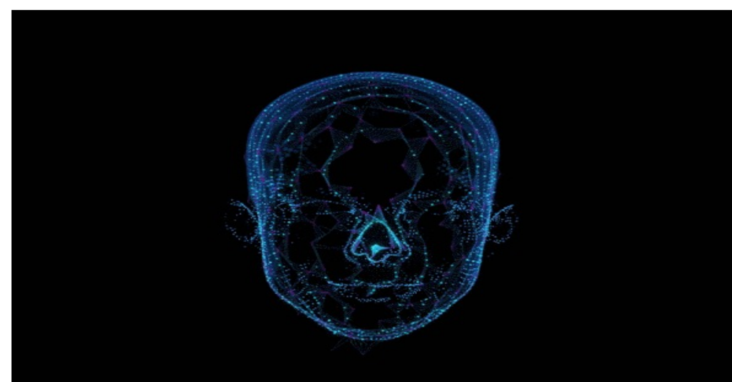
Playing chess



Purchasing suggestions on e-commerce site



Self-driving cars



Speech recognition and image recognition

AI type-1: Based on Capabilities

2. General AI:

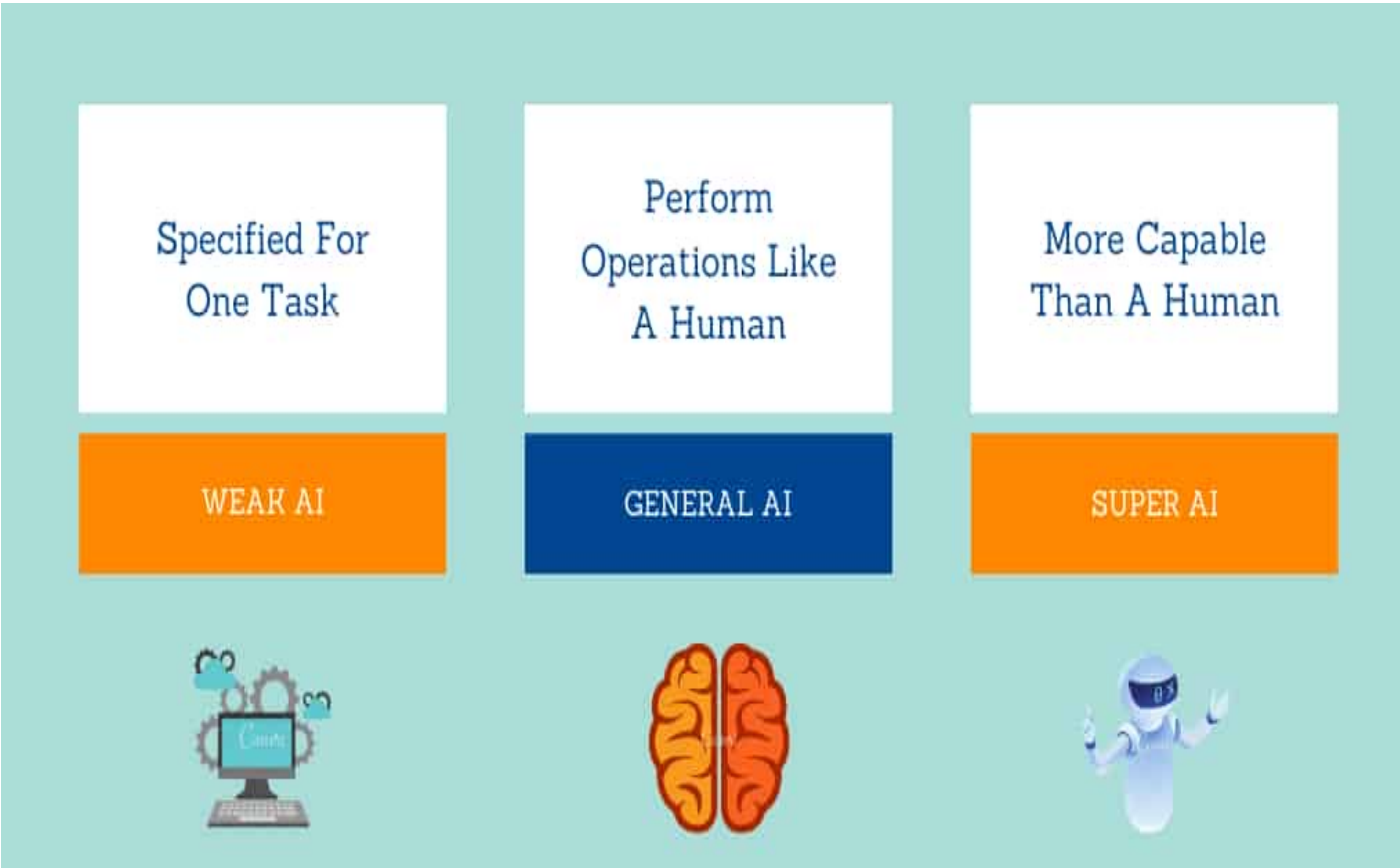
- General AI is a type of intelligence that can perform any intellectual task with efficiency like a human.
- The idea behind the general AI is to make such a system that could be smarter and think like a human on its own.
- Currently, there is no such system exists which could come under general AI and can perform any task as perfectly as a human.
- The worldwide researchers are now focused on developing machines with General AI.
- Systems with general AI are still under research, and it will take lots of effort and time to develop such systems.

Narrow AI	General AI
○ Application specific/ task limited	○ Perform general (human) intelligent action
○ Fixed domain models provided by programmers	○ Self-learns and reasons with its operating environment
○ Learns from thousands of labeled examples	○ Learns from few examples and/or from unstructured data
○ Reflexive tasks with no understanding	○ Full range of human cognitive abilities
○ Knowledge does not transfer to other domains or tasks	○ Leverages knowledge transfer to new domains and tasks
○ Today's AI	○ Future AI?

AI type-1: Based on Capabilities

3. Super/Strong AI

- Super AI is a level of Intelligence of Systems at which machines can surpass human intelligence, and can perform any task better than humans with cognitive properties.
- It is an outcome of general AI.
- Some key characteristics of strong AI include capability including the ability to think, reason, solve the puzzle, make judgments, plan, learn, and communicate on its own.
- Super AI is still a hypothetical concept of Artificial Intelligence. The development of such systems in real is still a world-changing task.


Specified For One Task
Perform Operations Like A Human
More Capable Than A Human
WEAK AI
GENERAL AI
SUPER AI

AI type-2: Based on functionalities

1. Reactive Machines (1/3):

- Purely reactive machines are the most basic types of Artificial Intelligence.
- Such AI systems do not store memories or past experiences for future actions.
- These machines only focus on current scenarios and react to them as per the possible best action.

AI type-2: Based on functionalities

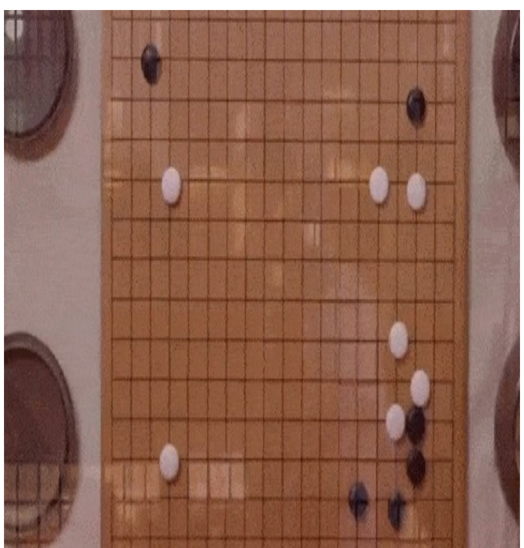
1. Reactive Machines (2/3):

- IBM's Deep Blue system is an example of a reactive machines.

AI type-2: Based on functionalities

1. Reactive Machines (3/3):

- AlphaGo learned to master the ancient Chinese game of Go — a profoundly complex board game of strategy, creativity, and ingenuity.



AlphaGo seals 4-1 victory over Go grandmaster Lee Sedol

DeepMind's artificial intelligence astonishes fans to defeat human opponent and offers evidence computer software has mastered a major challenge

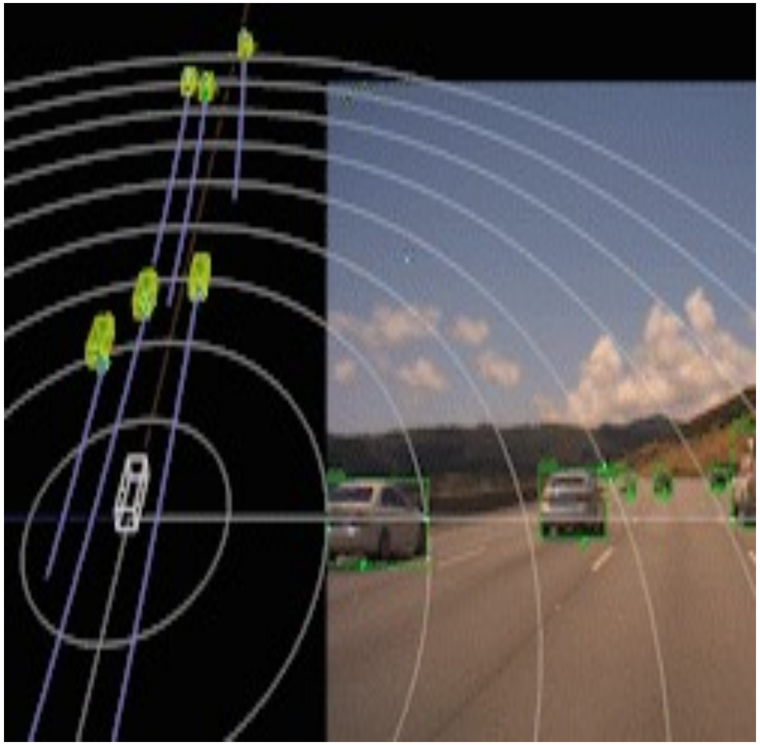
The world's top Go player, Lee Sedol, lost the final game of the Google DeepMind challenge match. Photograph: Yonhap/Reuters

Google DeepMind's AlphaGo program triumphed in its final game against South Korean Go grandmaster Lee Sedol to win the series 4-1, providing further evidence of the landmark achievement for an artificial intelligence program.

AI type-2: Based on functionalities

2. Limited Memory:

- Limited memory machines can store past experiences or some data for a short period.
- These machines can use stored data for a limited period only.
- Self-driving cars are one of the best examples of Limited Memory systems. These cars can store the recent speed of nearby cars, the distance of other cars, the speed limit, and other information to navigate the road.



AI type-2: Based on functionalities

3. Theory of Mind:

- Theory of Mind AI should understand human emotions, people, and beliefs, and be able to interact socially like humans.

- This type of AI machine is still not developed, but researchers are making lots of efforts and improvements to develop such AI machines.

AI type-2: Based on functionalities

4. Self-Awareness:

- Self-awareness AI is the future of Artificial Intelligence. These machines will be super intelligent and will have their consciousness, sentiments, and self-awareness.
- These machines will be smarter than the human mind.
- Self-awareness AI does not exist in reality still, and it is a hypothetical concept.

AI type-2: Based on functionalities

Reactive AI	Limited memory	Theory of mind	Self-aware
Good for simple classification and pattern recognition tasks	Can handle complex classification tasks	Able to understand human motives and reasoning. Can deliver personal experience to everyone based on their motives and needs.	Human-level intelligence that can bypass our intelligence, too
Great for scenarios where all parameters are known; can beat humans because it can make calculations much faster	Able to use historical data to make predictions	Able to learn with fewer examples because it understands motive and intent	
Incapable of dealing with scenarios including imperfect information or requiring historical understanding	Capable of complex tasks such as self-driving cars, but still vulnerable to outliers or adversarial examples	Considered the next milestone for AI's evolution	
	This is the current state of AI, and some say we have hit a wall		


THANK
YOU